

TILAK PARAJULI

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AI and Machine Learning researcher with hands-on experience in **Large Language Models (LLMs)**, **information retrieval**, **retrieval-augmented generation (RAG)**, and **data-driven research systems**. Proficient in **Python** and **ML/DL frameworks**, with applied expertise in **NLP**, **OCR**, **statistical modeling**, **biostatistics**, and **multimodal AI for healthcare and computational biology**. Passionate about **generative AI**, **agentic and autonomous systems**, and pathways to **general-purpose intelligence**. Self-driven and collaborative, with a proven track record in innovative research environments.

RESEARCH INTERESTS

Large Language Models (LLMs) & Reasoning, Retrieval-Augmented Generation (RAG), Hybrid Retrieval Systems, LLM Agents & Multi-Agent AI

Autonomy & Adaptive Intelligent Systems, Reinforcement Learning and Agentic RL, Self-Improving AI & AGI Pathways

Multimodal Perception & Understanding, Geometric and Relational Deep Learning, AI in Decision-Making Systems

Computational Biology & Genomics, Cancer Genomics & Progression, Epigenomics & Regulatory Networks, Biostatistics, Machine Learning in Biomarker Discovery & Therapeutic Resistance

Explainability, Interpretability & Robustness in AI, Ethical & Trustworthy AI, AI for Real-World Impact (Healthcare, Finance, Cybersecurity)

EDUCATION

Bhaktapur Multiple Campus, Tribhuvan University Dudhpati, Bhaktapur
Bachelor of Science in Computer Science and Information Technology — 78/100 Dec. 2019 – Oct. 2024
Major/Elective Courses: Data Structures and Algorithms | Geographical Information System | Probability and Statistics | Multimedia Computing | Computer Graphics | Data Mining | Artificial Intelligence | Information Retrieval

WORK EXPERIENCE

Data Scientist Jan 2026 – Present
Climate Clean Solutions and LowPropTax, Hybrid California, USA / Kathmandu, Nepal

- Developing and optimizing a **multi-agent AI system** for property tax appeals using **LLMs**, **RAG/CAG**, and API-driven comparable retrieval.
- Designing and refining **adjustment and valuation models** incorporating physical attributes, neighborhood signals (**crime**, **schools**), and **climate factors**.
- Implementing **validation and QA pipelines** with historical benchmarking, enforcing **>10% reduction thresholds** and automated **PDF/Excel** outputs.
- Building **real-time data pipelines** for air quality using **AirGradient sensors** and **OpenWeather APIs** (PM2.5, NO₂, O₃, CO₂).
- Managing and optimizing an **S3-based data lake** with **Parquet** storage, data cleaning, **statistical analysis**, and **dimensional modeling**.
- Developing **Power BI** dashboards for **time-series analysis**, **anomaly detection**, **geospatial insights**, and environmental-health correlations.

AI/ML Intern June 2025 – September
Fusemachines Nepal, Internship Kathmandu, Nepal

- Developed a **time-series model evaluation notebook** with metrics (**MAE**, **RMSE**, **MAPE**, **SMAPE**) and **visualizations**; refined and deployed based on peer review.

- Implemented **real-time gaze detection** in Fuse Proctor front-end using **MediaPipe FaceMesh** and **React**; enabled **directional gaze tracking** and **in-browser calibration**.
- Created **academic notebooks and prototypes** on **Agentic AI** concepts, including **agent loops**, **memory management**, and **prompt strategies**; supported integration planning.
- Explored **vision-language models (CLIP, BLIP, PaliGemma 2)** and collaborated on frontend improvements for **liveness detection** and **multi-person monitoring**.

RPA Developer

February 2025 – May 2025

Quickfox Consulting, Full Time

Kathmandu, Nepal

- Developed and deployed **RPA bots** using the **RPA Framework** to automate repetitive tasks across desktop and web environments.
- Automated **data entry, form submissions, web scraping, emailing, and Excel file merging**, significantly improving **efficiency, accuracy, and productivity**.
- Led a complex **data cleaning** project on legacy medical databases, leveraging **large language models (LLMs)** to standardize and validate noisy data.
- Collaborated with cross-functional teams to identify **automation opportunities** and ensured **bot reliability** through comprehensive testing and validation.

AI Research Intern

June 2024 – September 2024

NAAMII, Hybrid

Kathmandu, Nepal

- Contributed to a **radiation oncology** project by performing **clinical and radiomics data analysis**.
- Conducted **data preprocessing, normality testing, statistical tests** including **permutation tests, ROC AUC, and correlation analysis** on independent and dependent samples.
- Developed **statistical models and machine learning algorithms** under supervision of Dr. Taman Upadhaya.
- Applied **Statistical Analysis, Machine Learning, NLP, and Deep Learning** techniques in research activities.
- Demonstrated strong **commitment to innovation** and advancing AI research methodologies.

LEADERSHIP & ACTIVITIES

Student Union Member

March 2023 – Present

Bhaktapur Multiple Campus, Tribhuvan University

Bhaktapur, Nepal

- Elected with one of the **highest vote counts** in the **TU FSU 2023** election, representing students as a union member.
- Organized **campus-wide events**, advocated for **student welfare**, and served as a **liaison between students and faculty/administration**.

TECHNICAL SKILLS

Languages: Python, C, C++, Bash

Data Science & ML: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, MLflow

LLMs & Agentic AI: Hugging Face, LangChain, LangGraph, RAG, CAG

Data Engineering: Airflow, Parquet, ETL Pipelines, Dimensional Modeling

Cloud & MLOps: AWS (S3, EC2, IAM), Docker, Git, GitHub

BI & Visualization: Power BI, GIS Mapping, Time-Series Dashboards

Web & APIs: FastAPI, Django, RESTful APIs

Databases: MySQL, PostgreSQL, MongoDB

PROJECTS

RAG Pipeline for Job Data Retrieval

Jul. 2025

Python, FastAPI, LlamaIndex, ChromaDB, Cohere, BM25, Hybrid Retrieval, Docker

- Designed and implemented an end-to-end **Retrieval-Augmented Generation (RAG)** system for **intelligent job search and matching** over structured and unstructured job datasets.
- Built a **hybrid retrieval pipeline** combining **BM25 sparse retrieval** and **dense vector embeddings**, improving recall and semantic relevance.
- Integrated **LlamaIndex** for document ingestion, chunking, indexing, and query orchestration with configurable retrieval strategies.
- Implemented **Cohere embeddings** and **Cohere reranking** to enhance contextual relevance of retrieved job postings before generation.

- Developed a **FastAPI-based backend** exposing REST APIs for job queries, retrieval evaluation, and LLM-powered response generation.
- Engineered a **text preprocessing and metadata injection pipeline** to ensure semantically self-contained document chunks.
- Containerized the full system using **Docker** for reproducible deployment and scalable experimentation.

Travya - AI-Powered Travel Companion

Jan. 2025 – Dec. 2025

Python, FastAPI, React, TypeScript, PostgreSQL, Redis, Docker

- Developed an **AI-powered travel planning platform** using a **multi-agent architecture** with specialized agents for research, planning, and booking operations.
- Implemented a **real-time conversational AI system** with **streaming responses** and conversation context management using FastAPI and React.
- Designed and integrated a **RAG-based recommendation engine** with vector search for personalized travel suggestions.
- Integrated **external APIs** including Google Maps, Amadeus, Stripe, and weather services for end-to-end travel workflows.
- Designed a **microservices architecture** with PostgreSQL persistence, Redis caching, and Docker-based deployment.
- Implemented **JWT-based authentication**, role-based access control, and secure API endpoints.

Text Summarization with LSA and T5 Transformers

Feb. 2024 – Apr. 2024

Python, Transformers (T5), NLP, Machine Learning

- Implemented a **dual-approach text summarization system** using **Latent Semantic Analysis (LSA)** and **T5 transformers** for extractive and abstractive summarization.
- Engineered a **TF-IDF threshold-based feature extraction pipeline** for preview summaries in the UI.
- Evaluated summarization quality using **ROUGE metrics** across both classical and neural approaches.
- Built a robust **text preprocessing pipeline** including cleaning, tokenization, and feature engineering.

LICENSES AND CERTIFICATES

- **Full Scholarship – Sixth Annual Nepal AI School (ANAIS 2025)** Program Details *Awarded a full scholarship to attend an intensive 11-day Artificial Intelligence program featuring world-class lectures, hands-on labs, and project work with global AI researchers and practitioners.*
- **AI Fellow (2025) – Fusemachines AI Fellowship (Microdegree™)** Program Details — Certificate *Selected through a highly competitive, prestigious full scholarship program; an intensive 6-month curriculum covering Machine Learning, Deep Learning, NLP, Computer Vision, and MLOps, taught by global AI experts. Current Fellow.*
- **AI Agents and Agentic AI with Python & Generative AI** Certificate *Issued by: Vanderbilt University, Jul. 2025*
- **Level II Python Certification** Certificate *Issued by: Robocorp, Feb. 2025*
- **Programming in Python** Certificate *Issued by: Meta, Jan. 2024, Credential ID: 45GQSH2LGDZY*
- **Elements of AI** Certificate *Issued by: University of Helsinki and Reaktor, Mar. 2022*

REFERENCES

Taman Upadhaya, M.S., Ph.D. — *Researcher, Cedars-Sinai & NAAMII; Expert in Data Science, Computer Vision, and Medical Image Processing*

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